

QUITO, Ecuador

WMO: 840710

Lat: 0.141S

Lon: 78.488W

Elev: 2813

StdP: 71.79

Time Zone: -5.00 (W05)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.3	7.2	-0.1	5.3	15.1	1.1	5.8	14.7	8.5	16.9	7.8	17.0	0.5	340	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.4	21.9	11.8	21.1	11.9	20.8	11.9	14.5	18.8	14.1	18.3	13.7	17.9	3.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	13.3	16.1	12.5	12.8	15.3	12.2	12.5	14.9	51.8	19.0	50.4	18.5	49.4	18.2	17.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.7	6.9	6.1	DB	3.8	23.8	1.0	1.2	3.1	24.6	2.5	25.3	2.0	26.0	1.3	26.8	(4)
			WB	2.0	15.7	1.4	0.5	1.0	16.1	0.2	16.4	-0.6	16.7	-1.6	17.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	14.4	14.5	14.4	14.4	14.5	14.3	14.4	14.7	14.7	14.5	14.3	14.2
	DBStd	1.18	1.23	1.39	1.22	1.00	1.18	1.08	1.26	1.25	1.11	1.09	1.12	1.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1417	122	108	121	117	118	121	121	113	108	118	122	127
	CDD10.0	1626	137	126	138	133	140	129	137	145	142	141	128	131
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	0	0	0	1	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	2.1	2.0	2.0	1.8	1.7	1.8	2.2	2.6	2.8	2.5	2.0	1.8	1.9
Precipitation	PrecAvg	1083	88	107	134	161	106	49	26	29	73	115	107	85
	PrecMax	1453	170	233	280	291	171	125	146	78	131	223	254	125
	PrecMin	742	14	29	44	75	37	8	1	2	19	22	10	13
	PrecStd	217	42	64	50	58	39	27	32	26	34	53	65	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	22.1	21.7	21.1	21.8	21.1	21.7	22.0	22.2	22.1	21.2	21.1
		MCWB	12.5	12.6	12.6	12.9	12.8	11.8	11.2	10.9	11.0	11.6	12.4	12.7
	2%	DB	20.8	20.8	20.1	20.0	20.2	20.1	20.8	21.0	21.1	21.0	20.2	20.0
		MCWB	12.7	13.0	12.8	12.9	12.5	11.5	11.0	10.9	11.0	11.9	12.2	12.7
	5%	DB	19.8	19.8	19.2	19.1	19.8	19.2	19.9	20.1	20.2	20.0	19.2	19.1
		MCWB	12.7	12.8	12.8	13.1	12.5	11.5	10.9	10.9	11.0	12.0	12.4	12.7
	10%	DB	18.8	18.2	18.1	18.1	18.8	18.8	19.0	19.2	19.3	19.0	18.2	18.1
		MCWB	12.6	12.7	12.7	12.9	12.5	11.6	11.0	10.9	11.1	12.1	12.4	12.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	14.9	14.7	14.7	14.8	14.0	13.6	13.5	13.7	14.1	14.4	14.7
		MCDB	19.3	19.4	18.9	18.6	18.9	18.2	18.7	19.0	18.8	19.0	18.7	18.9
	2%	WB	14.0	14.1	14.0	14.1	14.0	13.4	12.7	12.6	13.0	13.5	13.7	13.9
		MCDB	18.4	18.4	18.1	18.1	18.1	17.7	17.8	18.2	18.3	18.4	18.1	18.0
	5%	WB	13.5	13.6	13.5	13.6	13.5	12.7	12.2	12.1	12.5	12.9	13.3	13.5
		MCDB	17.8	17.7	17.5	17.5	17.5	17.1	17.1	17.5	17.5	17.8	17.7	17.5
	10%	WB	13.0	13.1	13.0	13.2	13.0	12.3	11.7	11.6	11.9	12.5	12.7	12.9
		MCDB	17.1	17.0	16.9	16.9	17.0	16.6	16.6	16.9	16.7	16.9	16.8	16.7
Mean Daily Temperature Range	5% DB	MDBR	8.9	8.4	8.4	8.3	9.0	9.4	10.0	10.4	10.7	10.1	9.4	9.0
		MCDBR	10.8	10.2	10.0	9.7	10.6	10.5	11.0	11.2	11.8	11.8	10.8	10.4
		MCWBR	5.0	4.6	4.6	4.7	5.0	5.0	5.1	5.1	5.3	5.3	5.1	4.9
	5% WB	MCDBR	9.1	8.9	8.9	8.7	9.2	9.7	10.5	10.9	11.0	10.3	9.2	9.1
		MCWBR	4.5	4.4	4.4	4.5	4.7	4.9	5.0	5.2	5.1	4.9	4.5	4.5
Clear-Sky Solar Irradiance	taub	0.327	0.354	0.346	0.335	0.325	0.304	0.303	0.335	0.353	0.359	0.352	0.342	
	taud	2.534	2.433	2.476	2.519	2.547	2.621	2.616	2.474	2.416	2.417	2.446	2.485	
	Ebn at Noon	1003	976	975	965	951	959	965	952	954	959	970	983	
	Edh at Noon	108	121	116	108	101	92	93	111	121	122	117	112	
All-Sky Solar Radiation	RadAvg	3.71	3.52	3.48	3.62	3.57	3.63	3.84	4.06	4.06	4.00	3.88	3.62	
	RadStd	0.35	0.28	0.19	0.13	0.22	0.21	0.24	0.31	0.29	0.32	0.36	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page